OncoVision Project Report

Mammogram Image Analysis System for Breast Tumor Detection Using YOLOv8



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Dedication

To the soul of every patient who inspired me with the meaning of hope,

and to every researcher striving for a better tomorrow.

I dedicate this project with a sincere prayer that it may serve as a building block

added to the edifice of knowledge and service to humanity.

Chapter One: Planning Phase

Introduction : Project Goal and Expected Outcomes

- The OncoVision project aims to develop a web-based system capable of analyzing mammogram images to detect breast tumors using object detection models (YOLOv8). The system provides a user-friendly interface and a database to store results for performance tracking.

    - Expected Outcomes:

    - - Accelerating analysis time to just a few seconds per image.
    - - Improving detection accuracy for potential lesions.
    - - Providing reporting and statistical dashboards for progress tracking and performance evaluation.

Summary of Previous Studies,

- YOLO models (v3–v8) have been applied in medical imaging for breast tumor detection
- Research has employed visual augmentations and Roboflow to improve data quality.
- Comparison: R-CNN and RetinaNet deliver higher accuracy but require longer processing times;
  YOLOv8 balances speed and accuracy, making it more suitable for interactive web applications.
- demonstrating real-time detection capability with configurable confidence thresholds.

Project Community Description

Target Users:

- - Researchers
- - Computer science students
- - AI medical training labs
- - Radiologists (for training/research purposes)

Constraints and Priorities:

- - This system is strictly for research and educational use.
- - Ensuring data privacy and accurate labeling.
- - Ability to run locally without continuous internet access.

Supporting Data

- (Summaries of interviews with radiology experts/students, if available.)
- (Results of usability and accuracy surveys.)
- (Statistics collected from lab test sessions.)
- Note: Questionnaires and tools may be attached in the appendices.

Analysis Phase :Chapter Two

ERD Model and Database Analysis

SQLite (oncovision.db file) :Database

The core tables and their relationships were derived from the init_db.py script.

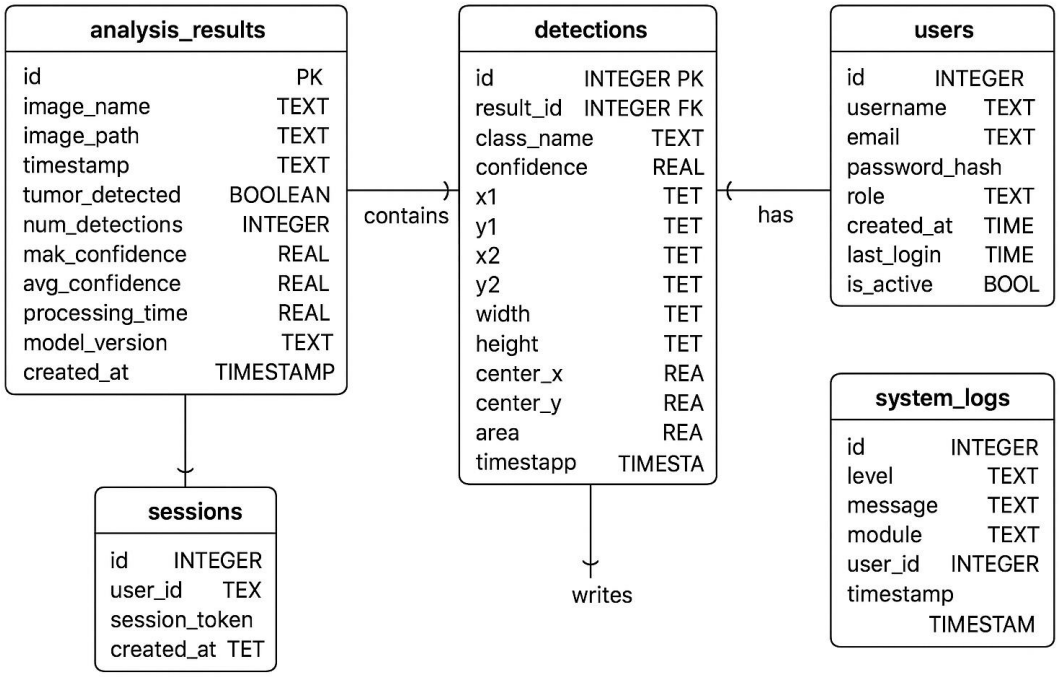


Table Descriptions (Summary):

- **analysis_results: Stores each analysis outcome, detection status, number of objects, max/average confidence, processing time, and model version.**
- **detections: Holds details of each detection (class, confidence, bounding box, center, area). The *area* field is auto-calculated.**
- **users: Contains user accounts with roles and status.**
- **sessions: Tracks user sessions and expiration times.**
- **system_logs: Stores system logs with level, module, and related user.**

Chapter Three: Design Phase

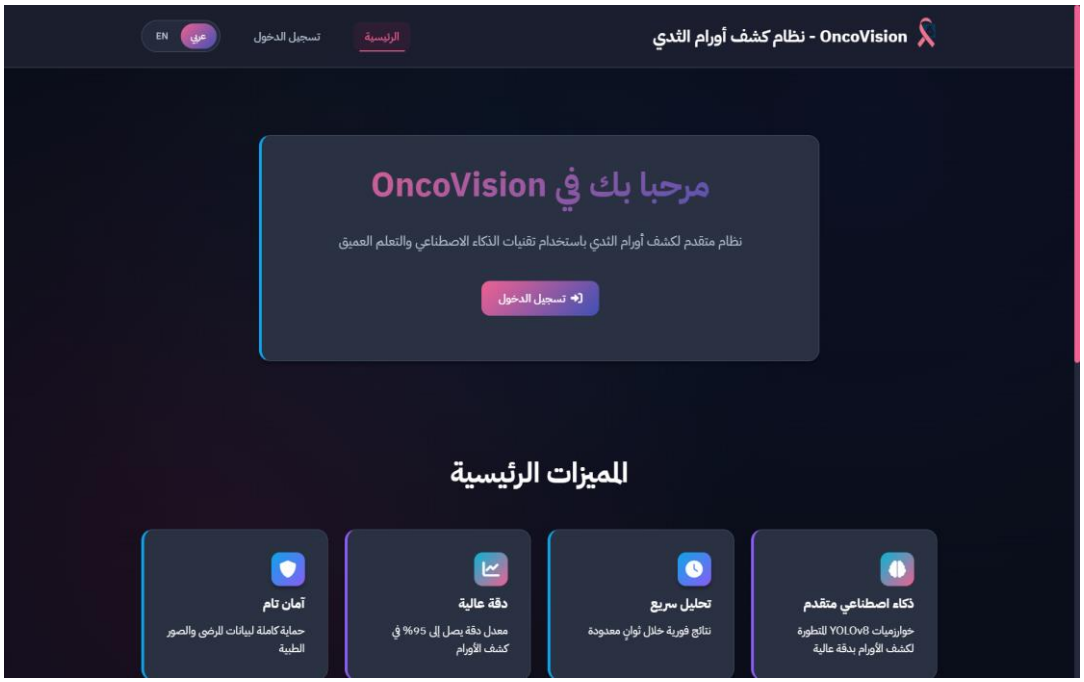
System Design

- **Layered Architecture:**
- **Presentation Layer**: Flask templates in `templates/` such as **index.html, login.html, analyze.html, dashboard.html, history.html, analysis_detail.html, statistics.html.**
- **Application Logic**: `app.py` (routes, sessions, authentication, file uploads, inference calls).
- **Inference Layer**: `infer.py` (OncoVisionInference class – wrapper for YOLOv8 model).
- **Data Layer**: SQLite via `init_db.py` with the mentioned tables.

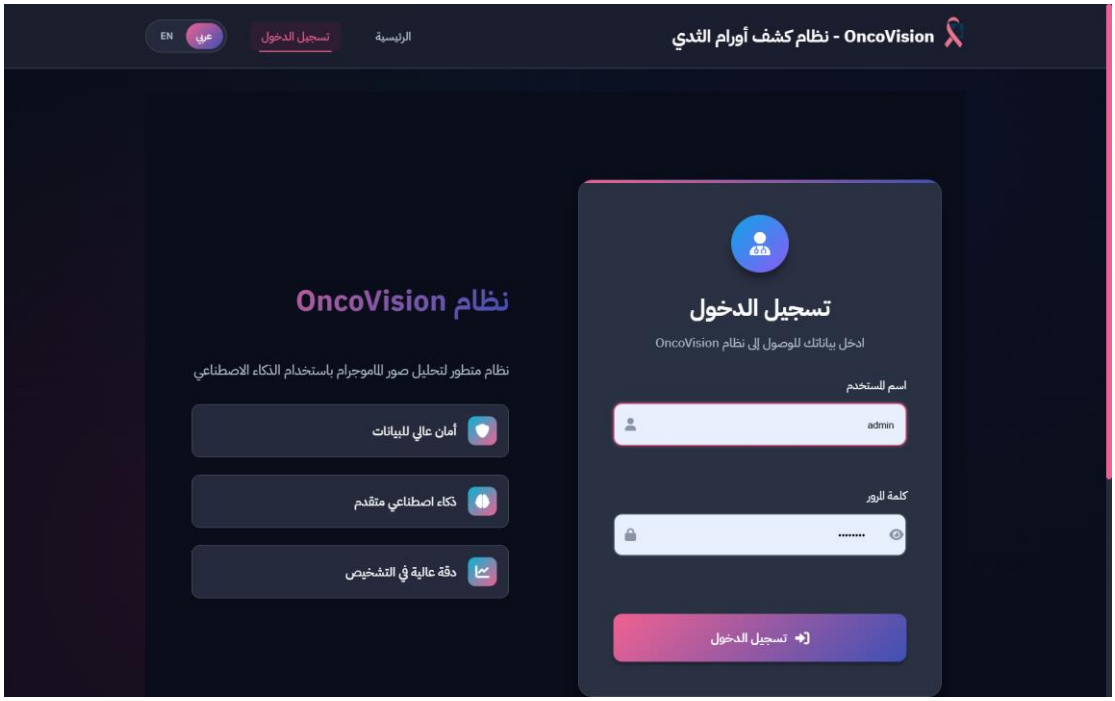
Forms/Models:

- Login, image upload, confidence & IoU parameters.
- **Reports:Dashboard, history, statistics pages pulling summaries from Views.**
- **Relationships:**Defined in the ERD diagram (Figure 2).

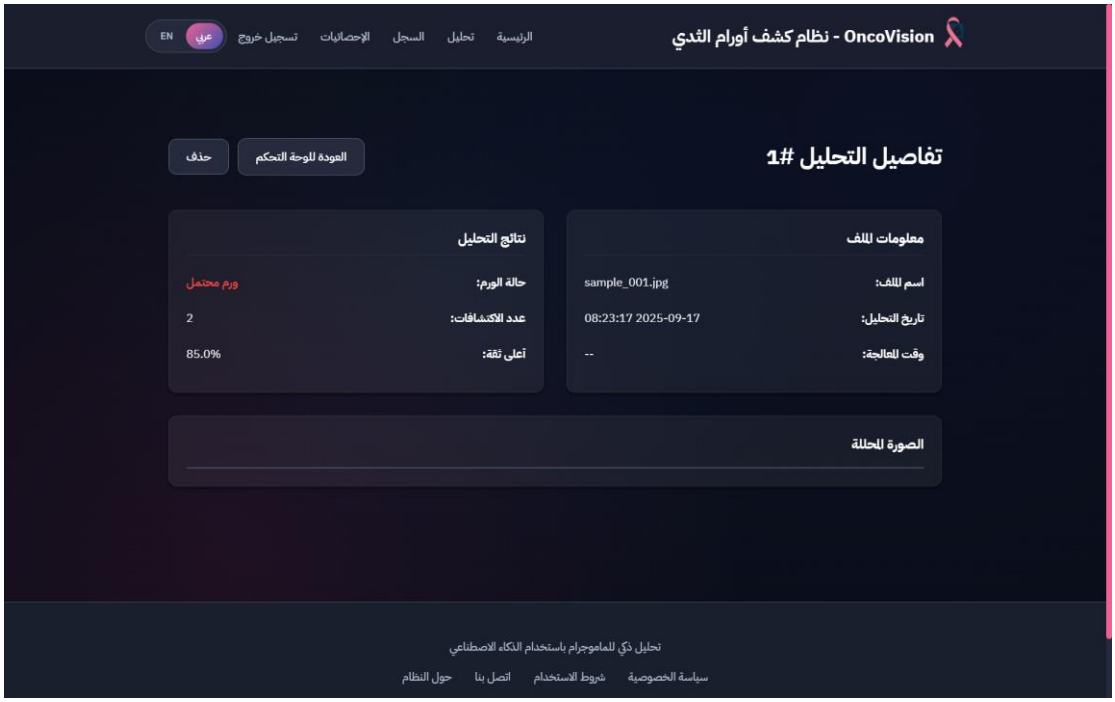
System Screenshots (placeholders in the report) •



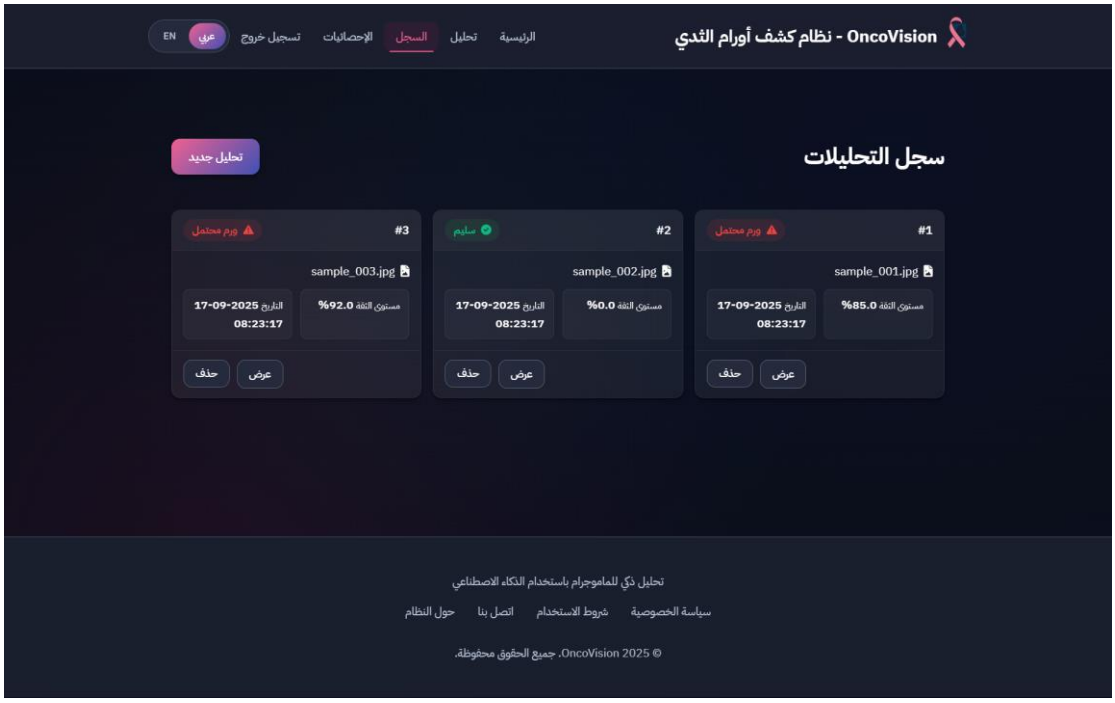
index.html — Start interface



login.html — Authentication screen

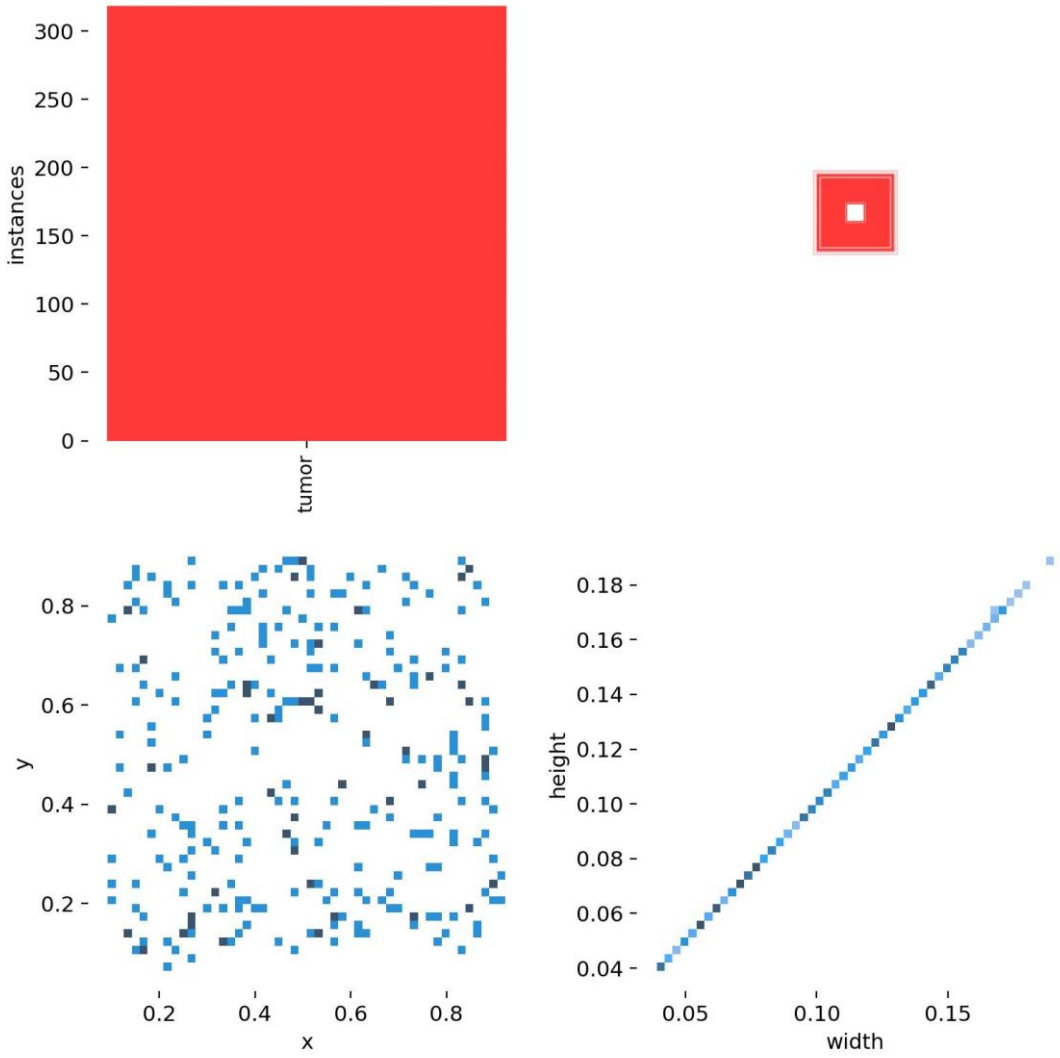


analyze.html — Upload & analysis results

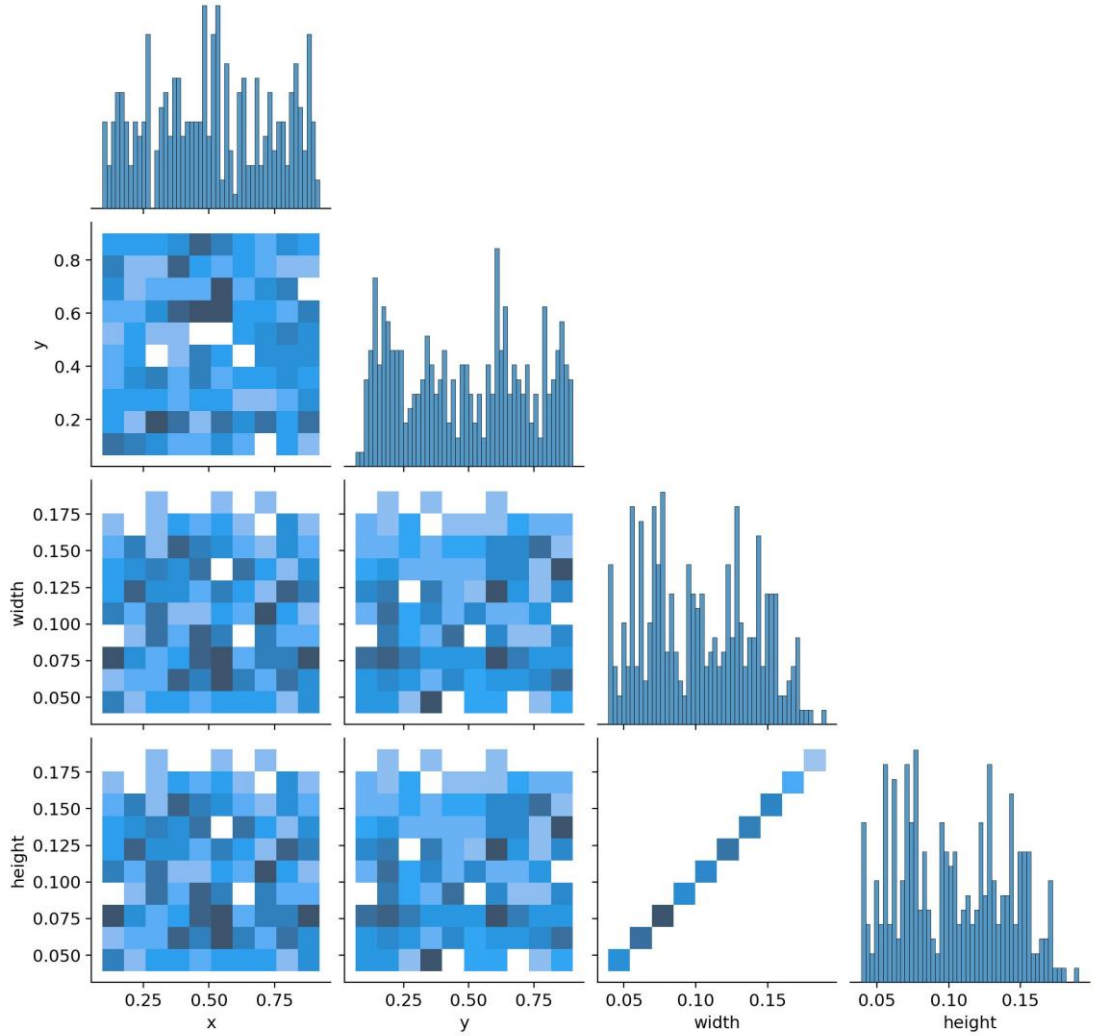


history.html — Analysis history with pagination

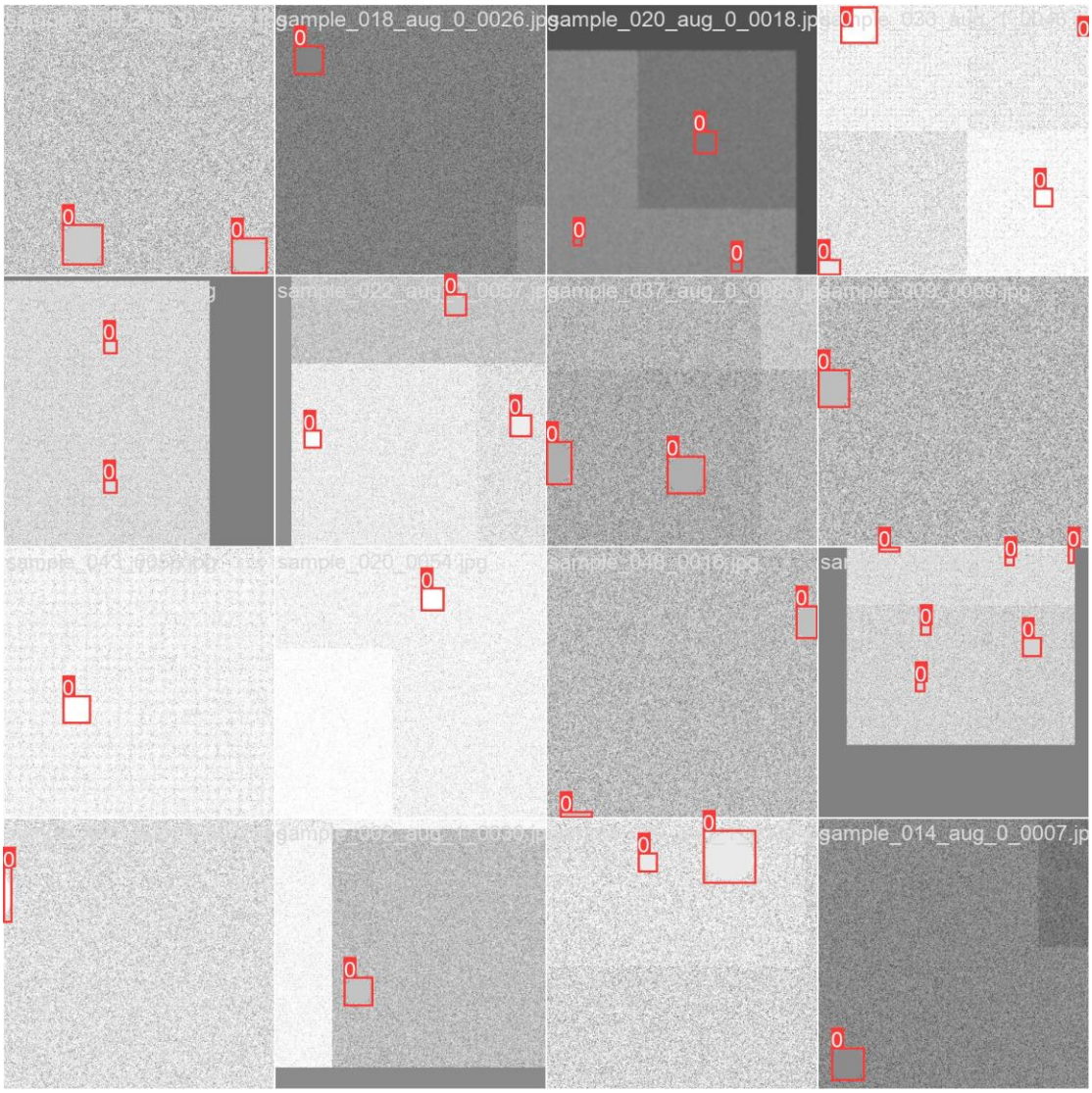
statistics.html — Statistics and charts



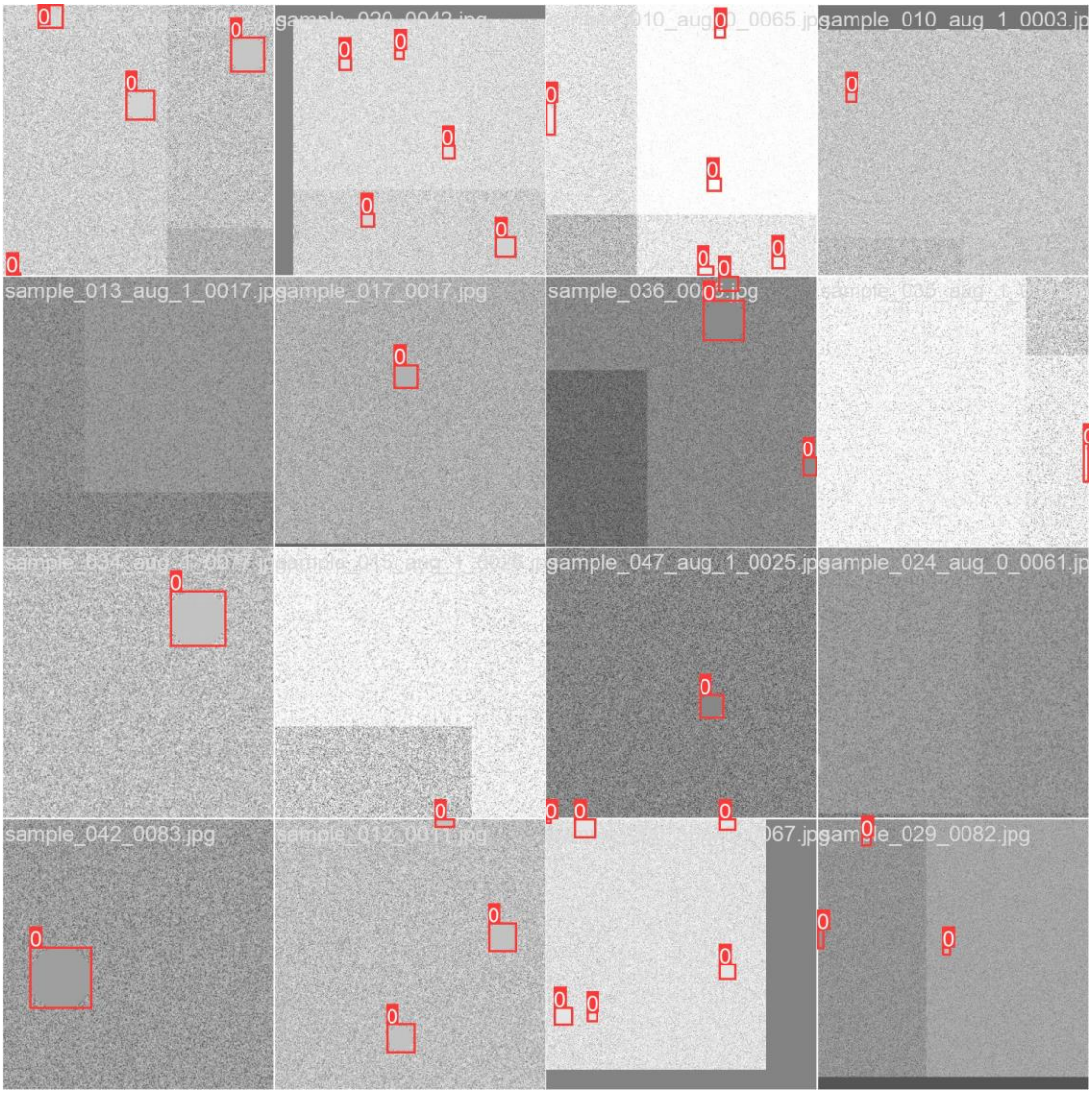
runs/oncovision_n_20250917_171442/labels.jpg



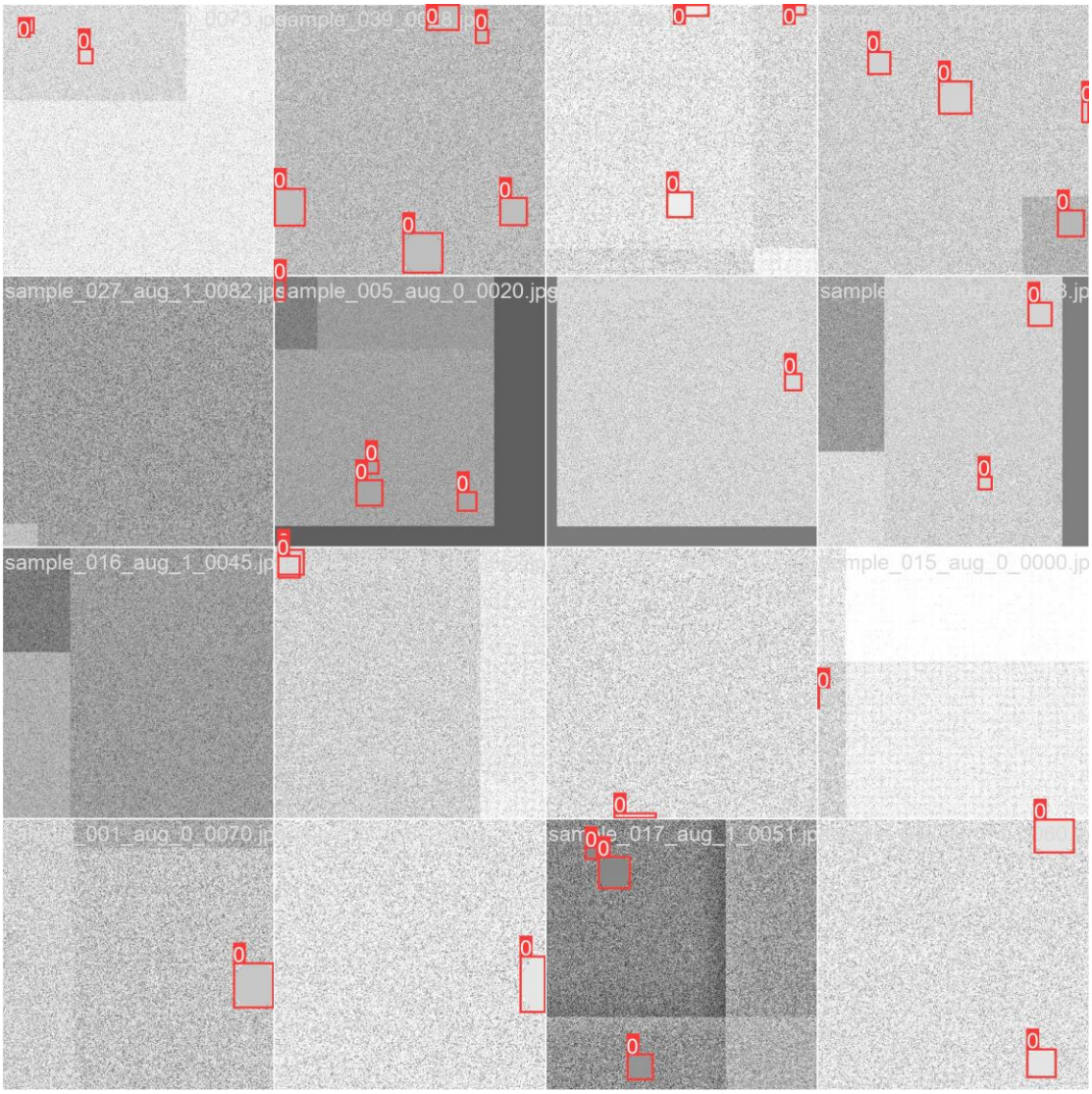
runs/oncovision_n_20250917_171442/labels_correlogram.jpg



runs/oncovision_n_20250917_171442/train_batch0.jpg



runs/oncovision_n_20250917_171442/train_batch1.jpg



runs/oncovision_n_20250917_171442/train_batch2.jpg

Chapter Four: Implementation Phase

Programming Languages and Technologies

- **Language**: Python 3.8+
- **Web Framework**: Flask 2.3.3
- **Computer Vision**: OpenCV 4.8.1.78, Pillow 10.0.0
- **Deep Learning**: PyTorch 2.0.1, Torchvision 0.15.2, Ultralytics YOLOv8 8.0.196
- **Data Analysis**: NumPy 1.24.3, Pandas 1.5.3, Seaborn 0.12.2, Matplotlib 3.7.2
- **Supporting Libraries/Tools**: tqdm, requests, PyYAML, Albumentations, Roboflow

Tools and Environments

- Development Environment: VS Code / Python virtual environment
- Version Control: Git (optional)
- Database: SQLite (oncovision.db)
- Data Management Scripts: prepare_yolo.py, train.py, evaluate.py, infer.py, init_db.py

YOLOv8 Data Preparation

- Config file: `data/data.yaml`
- Dataset path: `data/yolo_dataset`
- Sets: train/images, val/images, test/images
- Number of classes: nc = 1
- Class names: ['tumor']
- **Dataset sizes:**
- Training: 170 images / 170 labels
- Validation: 32 images / 32 labels
- Testing: 38 images / 38 labels

- **Application Security Note**



- **Passwords are stored as SHA-256 hashes.**
- **Sessions are managed via Flask.**
- **File upload size limited to 16MB.**

Chapter Five: Conclusion

Recommendations (Future Development)

- Support multiple classes (benign/malignant and additional regions).
- Add confidence calibration and explainability features (e.g., Grad-CAM).
- Enhance user and role management (Admin/Researcher/Student) with fine-grained permissions.
- Support integration with PACS or hospital systems through standard APIs.
- **Collect more data, expand visual augmentations, and improve class balance.**
- Improve application security in production: HTTPS, secrets management, secure file handling.

General Conclusion

- **OncoVision has presented a practical model for integrating YOLOv8 into an educational/research-oriented web application for mammogram analysis.**
  The system enables rapid testing, result tracking, and provides a foundation for wider experiments toward clinical environments, once medical and security requirements are met.

References

- Bochkovskiy, A., et al. YOLO: Real-Time Object Detection. (General YOLO reference)
- Ultralytics YOLOv8 Documentation — https://docs.ultralytics.com
- Breast Cancer Imaging Literature (relevant works)
- Flask Documentation — https://flask.palletsprojects.com
- PyTorch Documentation — https://pytorch.org
- PyTorch Documentation — https://pytorch.org

Appendices

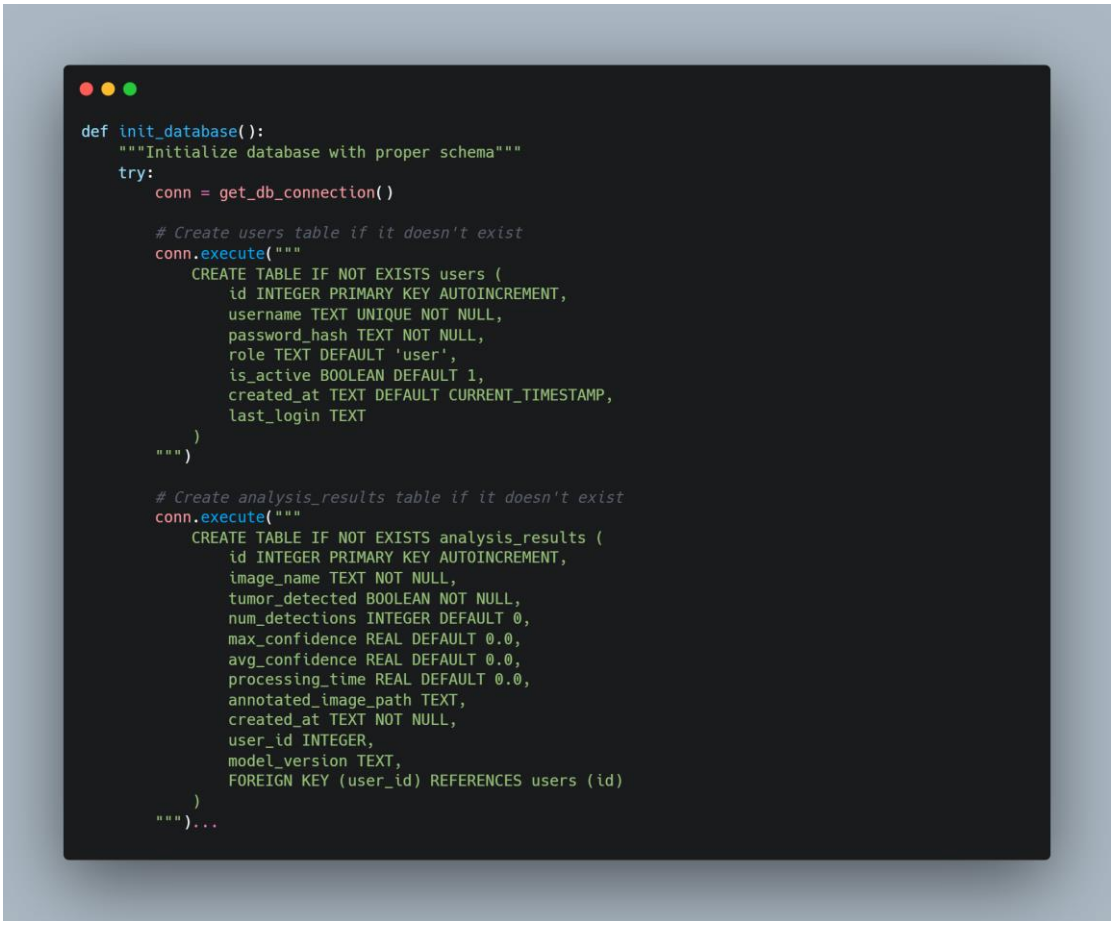
Appendix A: Questionnaires and Interviews

- Include texts of surveys, interview questions, and sample answers.

Appendix B: Code Snippets

- Example: Database initialization (init_db.py)
- Example: File upload and analysis route (app.py)
- Example: Data configuration file (data/data.yaml)

Database Initialization (init_db.py) :Excerpt



```python
def init_database():
    """Initialize database with proper schema"""
    try:
        conn = get_db_connection()

        # Create users table if it doesn't exist
        conn.execute("""
            CREATE TABLE IF NOT EXISTS users (
                id INTEGER PRIMARY KEY AUTOINCREMENT,
                username TEXT UNIQUE NOT NULL,
                password_hash TEXT NOT NULL,
                role TEXT DEFAULT 'user',
                is_active BOOLEAN DEFAULT 1,
                created_at TEXT DEFAULT CURRENT_TIMESTAMP,
                last_login TEXT
            )
        """)

        # Create analysis_results table if it doesn't exist
        conn.execute("""
            CREATE TABLE IF NOT EXISTS analysis_results (
                id INTEGER PRIMARY KEY AUTOINCREMENT,
                image_name TEXT NOT NULL,
                tumor_detected BOOLEAN NOT NULL,
                num_detections INTEGER DEFAULT 0,
                max_confidence REAL DEFAULT 0.0,
                avg_confidence REAL DEFAULT 0.0,
                processing_time REAL DEFAULT 0.0,
                annotated_image_path TEXT,
                created_at TEXT NOT NULL,
                user_id INTEGER,
                model_version TEXT,
                FOREIGN KEY (user_id) REFERENCES users (id)
            )
        """)...
```

Analysis Route and File Upload (app.py) :Excerpt

```python
@app.route('/analyze', methods=['GET', 'POST'])
@login_required
def analyze():
    """Analysis page"""
    if request.method == 'POST':
        if 'image' not in request.files:
            return jsonify({'success' : False, 'error' : 'No file selected'}), 400

        file = request.files['image']
        if file.filename == '':
            return jsonify({'success' : False, 'error' : 'No file selected'}), 400

        if not allowed_file(file.filename):
            return jsonify({'success' : False, 'error' : 'File type not supported. Please use: PNG, JPG,
JPEG, BMP, TIFF'}), 400

        try:
            # Save uploaded file
            filename = secure_filename(file.filename)
            timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
            unique_filename = f"{timestamp}_{filename}"
            filepath = app.config['UPLOAD_FOLDER'] / unique_filename
            file.save(filepath)

            # Get analysis parameters
            conf_threshold = float(request.form.get('confidence', 0.5))
            iou_threshold = float(request.form.get('iou', 0.45))

            start_time = datetime.now()...
```

Data Configuration File (data/data.yaml) :Excerpt

```yaml
# OncoVision Data Configuration
# YOLOv8 data configuration file

path: data/yolo_dataset  # Dataset root directory
train: train/images      # Training images directory
val: val/images          # Validation images directory
test: test/images         # Test images directory

# Number of classes
nc: 1

# Class names
names:
  0: tumor
```